

5(a)(i)

$$\begin{aligned}\frac{\text{slit width}}{\lambda} &= \frac{\text{slit width}}{\frac{v}{f}} \\ &= \frac{0.10 \times 10^{-3}}{\frac{3 \times 10^8}{4.8 \times 10^{14}}} \\ &= 160\end{aligned}$$

5(a)(ii)

wavelength is 625 nm, visible to human eye

each slit is a source of visible light of coherent phase difference

but slit width is not in same order of magnitude as wavelength of light,
little spreading of waves through each slit,

area on screen where the waves overlap is not wide

5(b)

$$\lambda = \frac{ax}{D}$$

$$x = \frac{\lambda D}{a} = \frac{vD}{fa}$$

$$= \frac{(3 \times 10^8)(2.6)}{(4.8 \times 10^{14})(1.5 \times 10^{-3})}$$
$$= 0.0011 \text{ m}$$

5(c)

fringe separation unchanged

bright fringe become dark fringe, dark fringe become bright fringe